

Implanted Electrical Stimulator, Spinal Cord

ACG: A-0243 (AC)
[Link to Codes](#)

This MCG care guideline may overlap CMS guidance located in CMS statutes and regulations that are not currently part of MCG's Medicare Compliance content. For Medicare Advantage beneficiaries, before referring to MCG care guidelines, users should first follow all guidance related to this topic located in CMS National Coverage Determinations (NCDs), Local Coverage Determinations (LCDs), and other statutes and regulations. Content used in MCG care guidelines may be used to supplement but does not replace, modify, or supersede existing Medicare regulations or applicable NCDs or LCDs.

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Clinical Indications

- Implanted electrical stimulator, spinal cord, may be indicated when **ALL** of the following are present(1):
 - Clinical condition is **1 or more** of the following:
 - Chronic neuropathic pain and **ALL** of the following(18)(19)(20)(21)(22):[N](#)
 - Condition appropriate for treatment, as indicated by **1 or more** of the following:
 - Complex regional pain syndrome (previously referred to as reflex sympathetic dystrophy)(33)
 - Diabetic neuropathy, refractory pain(30)
 - Failed back surgery syndrome[A](35)
 - Failed conservative treatment, as indicated by **1 or more** of the following:
 - Pharmacotherapy[B](37)
 - Physical therapy(37)
 - Psychotherapy or cognitive behavioral therapy
 - Stellate ganglion or lumbar sympathetic block
 - Pain due to critical limb ischemia and **ALL** of the following(38):[N](#)
 - Inoperable vascular disease or pain despite prior surgical or endovascular revascularization
 - Failed conservative treatment, as indicated by **1 or more** of the following:
 - Pharmacotherapy[B](37)
 - Physical therapy(37)
 - Psychotherapy or cognitive behavioral therapy
 - Planned procedure is **1 or more** of the following:
 - Percutaneous evaluation (ie, trial stimulation), if not previously performed(41)(42)
 - Permanent (implantable) stimulator placement, if percutaneous evaluation results in improvement in symptoms of 50% or greater(2)
 - Favorable psychological evaluation, absence of untreated psychiatric comorbidity, or current treatment in multidisciplinary pain management program(32)(43)(44)
 - Patient capable of operating stimulating device
 - No coagulopathy, anticoagulant or antiplatelet therapy, or thrombocytopenia (ie, platelet count of less than 75,000/mm³ (75 x10⁹/L))(44)(45)(46)

- No current or chronic infection(46)

Alternatives

- Alternatives include:
 - For complex regional pain syndrome(47)(48):
 - Cognitive behavioral therapy
 - Electrical nerve stimulation. See Electrical Nerve Stimulation, Transcutaneous (TENS) [↗](#) AC for further information.
 - Lumbar sympathetic block. See Nerve Block or Neurolysis, Lumbar Sympathetic [↗](#) AC for further information.
 - Pharmacotherapy, including possible intrathecal medication.(49)(50) See Intrathecal Pump Implantation [↗](#) AC for further information.
 - Physical therapy. See Pain Rehabilitation [↗](#) AC for further information.
 - Stellate ganglion block. See Nerve Block, Stellate Ganglion [↗](#) AC for further information.
 - For diabetic neuropathy, refractory pain(30):
 - Cognitive behavioral therapy
 - Pharmacotherapy
 - Physical therapy. See Pain Rehabilitation [↗](#) AC for further information.
 - For failed back surgery syndrome:
 - Cognitive behavioral therapy
 - Epidural corticosteroid injection.(15) See Epidural Corticosteroid Injection [↗](#) AC for further information.
 - Pharmacotherapy, including possible intrathecal medication.(50) See Intrathecal Pump Implantation [↗](#) AC for further information.
 - Physical therapy. See Spine Soft Tissue Dysfunction Rehabilitation [↗](#) AC for further information.
 - Spinal manipulation therapy. See Spinal Manipulation Therapy (SMT), Chiropractic and Other [↗](#) AC for further information.
 - For pain due to critical limb ischemia:
 - Cognitive behavioral therapy
 - Pharmacotherapy
 - Physical therapy. See Pain Rehabilitation [↗](#) AC for further information.

Evidence Summary

Background

Spinal cord stimulation involves implanting a spinal cord stimulator lead into the epidural space either percutaneously or by surgical laminectomy or laminotomy. Electrical impulses are generated by a pulse generator that is implanted subcutaneously. These electrical impulses generate stimuli that mask sensations of pain.(1) **(EG 2)** Prior to implantation of a permanent system, patients must undergo test screening with a percutaneous stimulation system to ensure that paresthesia is achieved.(2) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 39 published peer reviewed articles, 2 specialty society or other evidence-based guidelines, 3 Cochrane systematic reviews, and 1 book section.

For chronic neuropathic pain, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and meta-analysis of 8 randomized controlled trials (185 patients) evaluating spinal cord stimulation for the management of neuropathic pain found that spinal cord stimulation was associated with a reduction in pain intensity as measured by visual analog scale scores compared with controls. The authors noted that the small sample sizes, variable use of spinal cord stimulators, and differing control setups in the included studies may have limited the analysis.(23) **(EG 1)** A systematic review of 6 studies (including 5 randomized controlled trials) reported that spinal cord stimulation was effective for reducing chronic pain secondary to failed back surgery syndrome.(24) **(EG 1)** A randomized crossover trial of 50 patients with chronic radicular pain after lumbar spine surgery compared treatment with either 3 months of spinal cord stimulation or 3 months of sham stimulation and found, after 6 months of treatment, that there was no difference between groups in back pain disability as measured by the Oswestry Disability Index. The authors noted that further high-quality trials are necessary to determine the potential contribution of the placebo effect on patient outcomes for implanted spinal cord stimulators.(25) **(EG 1)** An observational study of 402 patients treated with a spinal cord stimulator for neuropathic pain (primary implantation or replacement implantation) reported a significant reduction in pain intensity at 2 years (defined as a 50% or greater reduction in pain intensity from baseline); however, only 77% of patients completed the 2-year follow-up. (26) **(EG 2)** A randomized open-label trial of 216 patients with medication-refractory diabetic neuropathy compared conventional medical management (eg, pharmacologic therapy, physical therapy, cognitive therapy, nerve blocks) with and without concomitant spinal cord stimulation and found, at 3-month follow-up, that those who received spinal cord stimulation were more likely to achieve at least 50% pain relief without worsening baseline neurologic deficits compared with conventional medical management alone.(27) **(EG**

1) A crossover extension of the previous study found, at 12-month follow-up, that 86% of patients who received spinal cord stimulation in the original treatment group achieved 50% or greater pain relief; among patients who crossed over from conventional medical management to spinal cord stimulation, 84% achieved 50% or greater pain relief after 6 months of follow-up. Additional follow-up is planned through 24 months.(28) **(EG 1)** A systematic review and meta-analysis of 2 randomized controlled trials (93 patients) evaluating the efficacy of spinal cord stimulation for the management of painful diabetic neuropathy found, after 6 months of follow-up, that spinal cord stimulation was associated with a greater proportion of patients achieving at least a 50% reduction in pain compared with best medical therapy; the authors noted that the analysis was limited by the small sample sizes and high risk of bias of the included studies. (29) **(EG 1)** A specialty society guideline suggests consideration of high-frequency spinal cord stimulation for patients with painful medication-refractory diabetic peripheral neuropathy.(30) **(EG 2)** Specialty society guidelines evaluating the appropriate use of spinal cord stimulation for the treatment of chronic pain recommend spinal cord stimulation for patients with failed back surgery syndrome and complex regional pain syndrome (previously referred to as reflex sympathetic dystrophy) who have failed conservative evidence-based interventions and are not candidates for corrective surgery.(1)(15)(18) **(EG 2)** A systematic review reported that psychological factors such as somatization, depression, anxiety, and poor coping skills are associated with poor outcomes for surgical interventions, including spinal cord stimulation.(31) **(EG 2)** Specialty society guidelines recommend assessment for high-risk psychosocial factors (eg, active psychosis, ongoing substance use disorder) prior to implantation of a spinal cord stimulator.(1)(32) **(EG 2)**

For pain due to critical limb ischemia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review that assessed the effectiveness of spinal cord stimulation for inoperable critical leg ischemia identified 6 studies (444 patients total) and found that, compared with patients who received conservative management (eg, analgesics, vasodilatory medication, anticoagulant medications, wound care), those who received spinal cord stimulation had better pain control and higher rates of limb salvage.(39) **(EG 1)** A specialty society guideline evaluating the appropriate use of spinal cord stimulation for the treatment of chronic pain and ischemic disease recommended consideration of spinal cord stimulation for patients with ischemic vascular disease who failed initial surgical or medical interventions for pain.(1) **(EG 2)** A specialty society guideline states that spinal cord stimulation may improve pain and amputation rates among patients with ischemic vascular disease. However, the authors note that the complication rate is high (17%) and that the treatment may not be cost-effective compared with conservative management.(40) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For angina pectoris (refractory), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review identified 7 randomized controlled trials with a total of 270 patients and concluded that spinal cord stimulation had efficacy and safety similar to percutaneous myocardial laser revascularization, making it a potential alternative treatment; however, the review stated that further high-quality trials were needed before stimulation could be accepted as a routine treatment for this condition.(3) **(EG 1)** A systematic review (538 patients total) concluded that spinal cord stimulation decreases anginal attacks and can improve functional status. Of note, in the 3 largest studies, 104 patients had 5-year follow-up, 104 patients had 2-year follow-up, and 68 patients had 1-year follow-up.(4) **(EG 2)** A study of 153 patients comparing enhanced external counterpulsation with spinal cord stimulation for refractory angina reported that, at 12-month follow-up, although both treatments reduced angina, enhanced external counterpulsation was slightly more effective than spinal cord stimulation.(5) **(EG 2)**

For cancer-related pain (refractory), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review concluded that current evidence is insufficient to establish the efficacy of spinal cord stimulation for refractory cancer-related pain.(6) **(EG 1)** A specialty society practice guideline states that spinal cord stimulation may be an option in patients with refractory cancer pain; however, the authors note that the clinical evidence is limited to case reports and case series.(7) **(EG 2)**

For chronic systolic heart failure, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A randomized, parallel, single-blind, controlled trial including 66 patients with New York Heart Association functional class III systolic heart failure who were implanted with a spinal cord stimulator found, at 6-month follow-up, that there was no significant difference in left ventricular end-systolic volume index, freedom from hospitalization for heart failure or death, or change in New York Heart Association functional classification between the active stimulation group and control group.(8) **(EG 1)**

For chronic visceral abdominal pain, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review of 7 studies (31 patients) evaluating spinal cord stimulation for the management of pain with chronic pancreatitis found that treatment with spinal cord stimulation was associated with a median 61% reduction in visual analog scale pain scores and a median 69% reduction in opioid use at the end of the follow-up period. However, the authors noted that the small sample sizes and lack of randomized controlled trials limited the analysis, and they recommended additional trials to assess for efficacy.(9) **(EG 1)** A national survey of pain management physicians was conducted due to a lack of consensus regarding patient selection and appropriate lead placement of spinal cord stimulation for this condition. Twenty-three physicians who treated 70 patients responded to the survey; the most frequent indications for treatment were pain due to chronic pancreatitis, postoperative abdominal adhesions, and gastroparesis. At a mean follow-up of 84 weeks for the 66 patients who received permanent implantation, spinal cord stimulation was associated with a reduction in pain scores and opioid use. The authors acknowledged that selection bias was a study weakness (ie, interventional pain management physicians with negative results may not have responded to the survey).(10) **(EG 2)**

For nonoperative lower back pain, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review and meta-analysis of 13 randomized controlled trials (699 patients) that compared the long-term (longer than 12-month) effects of implanted spinal cord stimulation with standard medical care for nonoperative low back pain found, with moderate-certainty evidence, that spinal cord stimulation did not improve back pain, leg pain, function, or quality of life compared with standard care. The authors noted that the risk of bias in the included trials was high and that adverse events were poorly reported.(11) **(EG 1)** A systematic review of 10 studies (357 surgery-naïve patients) evaluating the efficacy of spinal cord stimulation for the treatment of chronic back pain found that many studies demonstrated improvement in pain and function, but that the data were limited regarding patient satisfaction, safety, and comparative effectiveness against other therapies.(12) **(EG 2)** An industry-funded randomized controlled trial of 270 patients with inoperable back pain (confirmed by lack of identifiable surgical pathology on MRI) compared treatment with either conservative medical management (eg, physical therapy, medication optimization, interventional therapies) or spinal cord stimulation and found, at 6-month follow-up, that spinal cord stimulation was associated with an improved response rate (defined as 50% or more reduction in pain from baseline as measured by numeric rating scale score) compared with conservative management, with a rate of procedure-related adverse events of 8.6%. The authors noted that 24-month follow-up studies are planned to evaluate long-term outcomes.(13) **(EG 1)** A specialty society consensus panel noted that referral for spinal cord stimulator was generally inappropriate for patients with chronic low back pain who had not undergone surgery.(14) **(EG 2)** A specialty society guideline states that implanted spinal cord stimulation may be considered for selected patients with nonoperative low back pain but that the net benefit of the intervention is likely small.(15) **(EG 2)**

For postherpetic neuralgia, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A systematic review evaluating interventional treatments for postherpetic neuralgia included 6 case series (94 patients) of spinal cord stimulation and found that some patients benefit; further study was recommended to evaluate which subset of patients was most likely to respond to treatment.(16) **(EG 1)** A randomized controlled trial of 63 patients with herpes zoster-related pain (duration of 1 to 6 months) treated with either spinal cord stimulation or pulsed radiofrequency found that both groups experienced a decrease in numeric rating scale pain scores with no difference noted between the groups.(17) **(EG 1)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(51); 42 CFR 422.101(52)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(53); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(54); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(55)

Medicare Coverage Determinations: Medicare Coverage Database(56)

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Footnotes

[A] Failed back surgery syndrome is described as persistent low back pain, either with or without lumbosacral radiculopathy, after one or more spine surgeries.(34) [A in Context Link 1]

[B] There is some evidence to support the short-term benefit of opioids for mild to moderate pain, but the evidence for improvement in function is inconsistent. Given that chronic opioid therapy for noncancer pain often begins with acute opioid prescriptions, clinicians should provide the lowest effective dose for the shortest duration necessary (eg, 2 to 3 days) to alleviate pain when giving prescriptions for acute pain; limiting the duration of opioid therapy can minimize the need to taper to prevent withdrawal symptoms at the end of the course of opioids and limit unused opioids.(36) [B in Context Link 1, 2]

Codes

CPT®: 0784T, 0785T, 0788T, 0789T, 63650, 63655, 63661, 63662, 63663, 63664, 63685, 63688

HCPCS: C1767, C1778, C1816, C1820, C1822, C1823, C1826, C1827, C1883, C1897, L8678, L8679, L8680, L8681, L8682, L8683, L8685, L8686, L8687, L8688, L8689, L8695 [Hide]

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